Artificial Intelligence
Lecture 3: State Space Search

Question from Module 1: Lecture 1

- 1. Define intelligence.

Intelligence is a rather hard to define term.

Intelligence is often defined in terms of what we understand as intelligence in humans.

Allen Newell defines intelligence as the ability to bring all the knowledge a system has at its disposal to bear in the solution of a problem.

A more practical definition that has been used in the context of building artificial systems with intelligence is to perform better on tasks that humans currently do better.

Question from Lecture 1

2. What are the different approaches in defining artificial intelligence ?

1. Thinking rationally
2. Acting rationally
3. Thinking like a human
4. Acting like a human

Question from Lecture 1

3. Suppose you design a machine to pass the Turing test. What are the capabilities such a machine must have ?

1. Natural language processing
2. Knowledge representation
3. Automation reasoning
4. Machine learning

Question from Lecture 1

5. Design ten question to pose to a man/machine that is taking the Turing test.
6. Will building an artificially intelligent computer automatically shed light on the nature of natural intelligence?
7. List 5 tasks that you will like a computer to be able to do within the next 5 years.
8. List 5 tasks that computers are unlikely to be able to do in the next 10 years.

Question from Module 1: Lecture 2

1. Define an agent.

   An agent is anything that can be viewed as perceiving its environment through sensors and executing actions using actuators.

2. What is a rational agent ?

   A rational agent always selects an action based on the percept sequence it has received so as to maximize its (expected) performance measure given the percepts it has received and the knowledge possessed by it.

Question from Module 1: Lecture 2

3. What is bounded rationally ?

   A rational agent that can use only bounded resources cannot exhibit the optimal behaviour. A bounded rational agent does the best possible job of selecting good action given its goal, and given its bounded resources.

4. What is an autonomous agent ?

   Autonomous agents are software entities that are capable of independent action in dynamic, unpredictable environments. An autonomous agent can learn and adapt to a new environmrnt.

Question from Module 1: Lecture 2

5. Describe the salient features of an agent.

- An agent perceives its environment using sensors
- An agent takes actions in the environment using actuators
- A rational agent acts so as to reach its goal, or to maximize its utility
- Reactive agents decide their action on the basis of their current state and the percepts. Deliberative agents reason about their goals to decide their action.

Questions : Lecture 2

6. Find out about the Mars rover.

a) What are the percepts for this agent?
b) Characterize the operating environment.
c) What are the actions the agent can take?
d) How can one evaluate the performance of the agent?
e) What sort of agent architecture do you think is most suitable for this agent?

Mars rover (Lecture 2)

a) What are the percepts for this agent?

Sprit’s sensor include

a) Panoramic and microscopic cameras,
b) A radio receiver,
c) Spectrometers for studying rock samples including an alpha particle x-ray spectrometer, M”ossbauer spectrometer, and miniature thermal emission spectrometer

Mars rover (Lecture 2)

a) Characterize the operating environment

The environment (the Martian surface)

- Partially observable,
- Non-deterministic,
- Sequential,
- Dynamic,
- Continuous, and
- May be single-agent.

  If a rover must cooperate with its mother ship or other rovers, or mischievous Martians tamper with its progress, then the environment gains additional agents.

Questions (Lecture 2)

C) What are the actions the agent can take ?

This rover Sprit has

a) Motor-driven wheels for locomotion
b) Along with a robotic arm to bring sensors close to interesting rocks and a
c) Rock abrasion tool (RAT) capable of efficiently drilling 45mm holes in hard volcanic rock.
d) Sprit also has a radio transmitter for communication.

Autonomous Mars rover

d) How can one evaluate the performance of the agent ?

Performance measure: A Mars rover may be tasked with

1. Maximizing the distance or variety of terrain it traverses.
2. Or with collecting as many samples as possible.
3. Or with finding life (for which it receives 1 point if it succeeds and 0 points if it fails).

Criteria such as maximizing lifetime or minimizing power consumption are (at best) derived from more fundamental goals; e.g., if it crashed or runs out power in the field, then it can’t explore.

Mars Rover

e) What sort of agent architecture do you think is most suitable for this agent ?

Model-based reflex agent for low level navigation;

For route planning, experimentation etc, some combination of goal-based, and utility-based would be needed

Internet shopping agent

Internet book-shopping agent

a) Sensor: Ability to parse Web pages, interface for user requests
b) Environment: internet partially observable, partly deterministic, sequential, party static, discrete, single-agent (exception: auctions)
c) Actuators: Ability to follow links, fill in forms, display info to user
d) Performance Measure: Obtains requested books, minimizes cost/time
e) Agent architecture: goal based agent with utilities fro open-ended situations

Artificial Intelligence

Module 2: Problem Solving Using Search

Lecture 3: Introduction to state Space Search

Instructional Objectives for this module

- The student should be familiar with the following algorithms, and should be able to code the algorithms

❑ Greedy search

❑ DFS

❑ BFS

❑ Uniform cost search

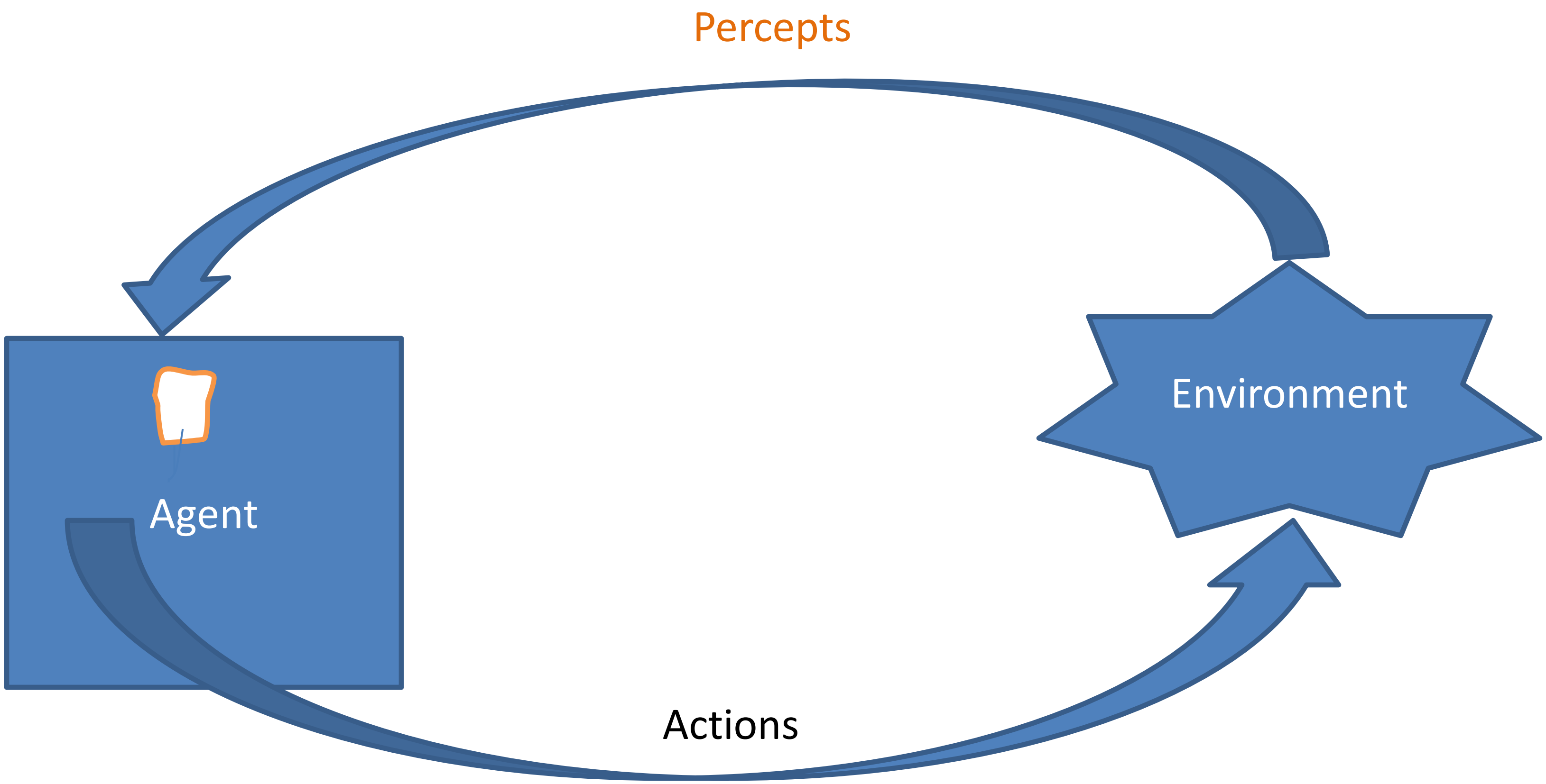
❑ Iterative deepening search

❑ Bidirectional search

Instructional Objectives for this lecture

- The students should understand the state space representation, and gain familiarity with some common problems formulated as state space search problems.
- Given a problem description, the student should be able to formulate it in terms of a state space search problem.
- The student should understand how implicit state spaces can be unfolded during search.
- Understand how sates can be represented by features.

Intelligent Agent



Goal directed Agent

- A goal directed agent needs to achieve certain goals.
- Many problems can be represented as a set of states and a set of rules of how one state is transformed to another
- The agent must choose a sequence of actions to achieve the desired goal.

Each state is an abstract representation of the agent's environment. It is an abstraction that denotes a configuration of the agent.

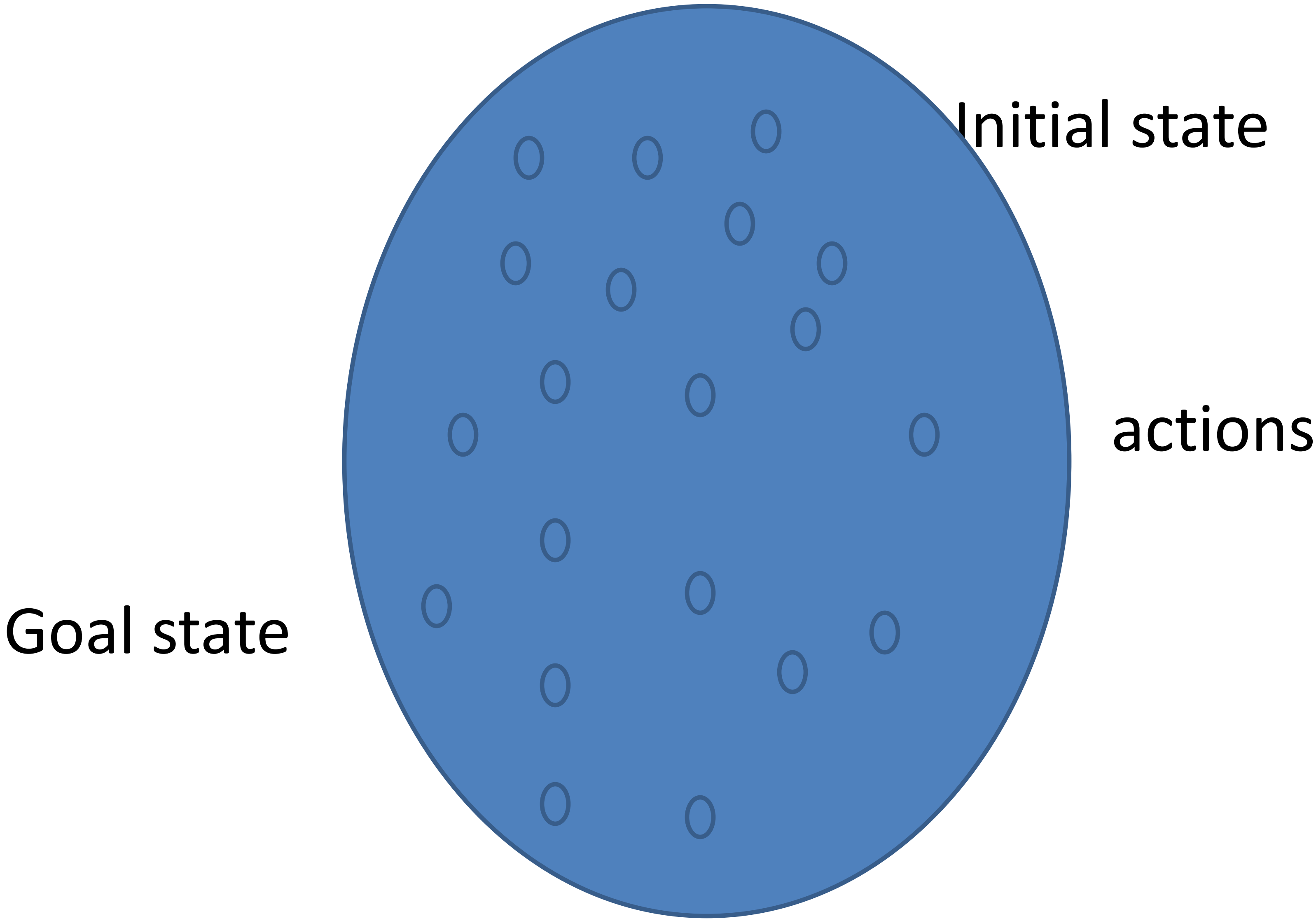
- Initial state: The description of the starting configuration of the agent
- An action/ operator takes the agent from one state to another state. A state can have a number of successor states.
- A plan is a sequence of actions.
- A goal is a description of set of desirable states of the world. Goal states are often specified by a goal test which any goal state must satisfy.

- Path cost: path - positive number
  usually path cost = sum of step costs
  plan-path
- Problem formulation means choosing a relevant set of states to consider, and a feasible set of operators for moving from one state to another.
- Search is the process of imagining sequences of operators applied to the initial state, and checking which sequence reaches a goal state.

Search problem

- S: the full of states
- S_0: Initial state
- A: S-S set of operators
- G: the set of final states.
- Search problem: Find a sequence of actions which transforms the agent from the initial state to a goal state a€G

State space



Searching process

- Check the current state
- Execute allowable actions to move to the next state
- Check if the new state is a solution state

  If it is not, the new state becomes the current state and the process is repeated until a solution is found or the state space is exhausted.

Pegs and Disk

- Consider the following problem. We have 3 pegs and 3 disks.
- Operators: one may move the topmost disk on any needle to the topmost position to any other needle

8 queens

- Place 8 queens on a chessboard so that no two queens are in the same row.

  Column or diagonal

N queens problem formulation 1

- States: any arrangement of 0 to 8 queens on the bard
- Initial state: 0 queens on the board
- Successor function: Add a queen in any square
- Goal test: 8 queens on the board. None are attacked

N queens problem formulation 2

- States: any arrangement of 8 queens on the bard
- Initial state: all queens are at column 1
- Successor function: change the position of any one queen
- Goal test: 8 queens on the board. None are attacked

N queens problem formulation 3

- States: any arrangement of k queens in the first k rows such that none are attacked
- Initial state: 0 queens on the board
- Successor function: Add a queen to the (k+1)th row so that none are attacked.
- Goal test: 8 queens on the board. None are attacked

Explicit vs Implicit state space

- The state space may be explicitly represented
- Typically it is implicitly represented and generated when required.
- The agent knows

  the initial state

  the operators.
- An operator is a function which “expands” a node

  compute the successor node(s)

Problem Definition-Example, 8 puzzle

5	**4**	
6	1	8
7	3	2

Initial state

1	**4**	**7**
2	5	8
3	6	

Goal state

Problem Definition-Example, 8 puzzle

- States

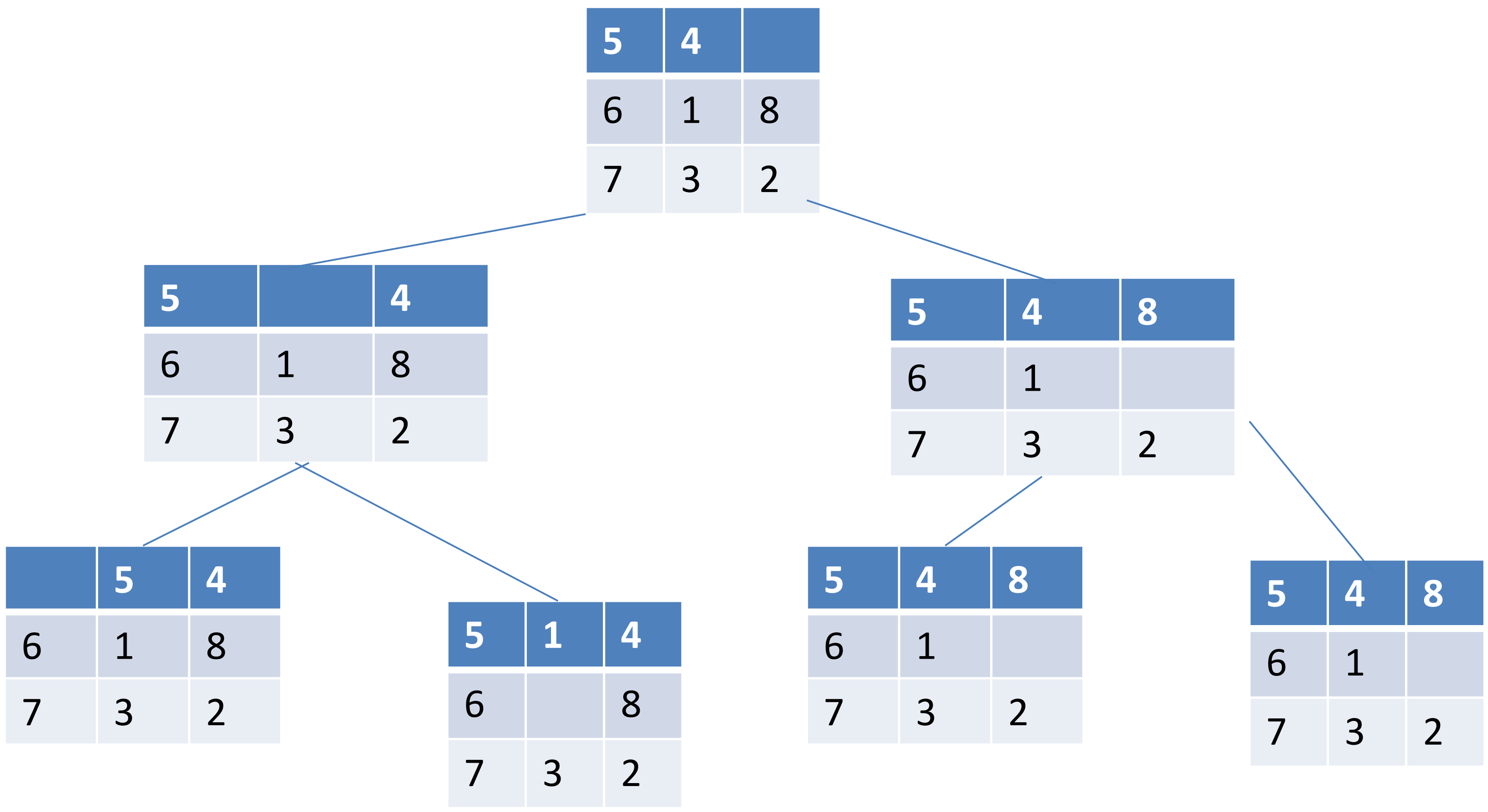
  A description of each of the eight tiles in each location that it can occupy.
- Operators/Action

  The blank moves left, right, up or down
- Goal test

  The current state matches a certain state (e.g. one of the ones shown on previous slide)
- Path cost

  Each move of the blank costs 1

Problem Definition-Example, 8 puzzle



Search through a state space

- Input

  Set of States

  Operators [and costs]

  Start state

  Goal state [test]
- Output

  Path : start- a state satisfying goal test

  [May require shortest path]

Basic search algorithm

Let L be a list containing the initial state (L=the fringe)
Loop
    if L is empty return failure
    Node select (L)
    if Node is goal
        then return Node
        (the path from initial state to Node)
    else apply all applicable operators to Node,
  and
    merge the newly generated sates into L

Basic search algorithm: Key issues

- Search tree may be unbounded
  - Because of loops
  - Because state space is infinite
  - Return a path or a node ?
  - How are merge and select done ?
  - Is the graph weighted or unweighted ?
  - How much is known about the quality of intermediate states?
  - Is the aim to find a minimal cost path or any path as soon as possible?

Search strategy

- Measuring problem solving performance:

  Completeness: Is the strategy guaranteed to find a solution if one exists?

  Optimality: Does the solution have low cost or the minimal cost?

  What is the search cost associated with time and memory required to find a solution?

Search strategy

- Blind search
  Depth first search
  Breadth first search
  Iterative deepening search
  Iterative broadening search
- Informed search
- Constraint Satisfaction
- Adversary search

Depth first search

Let L be a list containing the initial state (L=the fringe)

Loop

if L is empty return failure

Node select (L)

if Node is goal

then return Node the path from initial state to Node else apply all applicable operators to Node, and merge the newly generated sates into L

End Loop

Water jug problem

You have three jugs measuring 12 gallons, 8 gallons, and 3 gallons, and a water faucet. You need to measure out exactly one gallon.

Initial state: All three jugs are empty

Goal test: Some jug contains exactly one gallon.

Question for lecture 3

- Give the initial state, goal test, sucessor function, and cost function for each of the following.
- Choose a formulation that is precise enough to be implemented.

1. You have to colour a planar map using only four colours, in such a way that no two adjacent regions have the same colour.
2. In the travelling salesperson problem(TSP) there is a map involving N cities some of which are connected by roads. The aim is to find the shortest tour that starts from a city, visits all the cities exactly once and comes back to the starting city.